

Network Traffic Analysis Using Wireshark

Investigation of Suspicious Network Activity (Task #1003)

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Revision History

Version	Date	Author	Description
1.0	July 2026	Shirley Inocénté	Initial investigation report

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Executive Summary

This report documents the analysis of the Task #1003 packet capture (PCAP) using Wireshark to determine whether the captured network activity contained indicators of compromise (IoCs) or evidence of malicious behavior. The investigation followed a structured network traffic analysis methodology consisting of packet capture review, protocol analysis, packet-level inspection, and external threat intelligence validation.

The analysis examined multiple areas of network communication, including DNS queries, TLS-encrypted traffic, QUIC communications, SMB activity, and domain reputation. While some activity required additional review due to unfamiliar domains and encrypted communications, packet-level investigation and threat intelligence validation confirmed that the reviewed activity was associated with legitimate services and did not demonstrate indicators of malicious behavior.

The investigation did not identify confirmed malicious activity, unauthorized access attempts, suspicious data transfers, or communication with known malicious infrastructure. The findings demonstrate the importance of evidence-based analysis, where security analysts validate suspicious observations before determining whether activity represents a genuine security threat.

1. Introduction

Network traffic analysis is a critical component of cybersecurity operations, enabling security professionals to monitor network activity, investigate suspicious behavior, and identify potential indicators of compromise (IoCs). By examining packet-level data, analysts can reconstruct network conversations, validate the legitimacy of network communications, and detect behaviors that may indicate malicious intent.

Wireshark is one of the industry's most widely used network protocol analyzers because it provides detailed packet capture analysis and protocol decoding capabilities, allowing security professionals to examine network communications, protocols, hosts, and traffic patterns. During incident response, packet analysis helps security analysts distinguish legitimate network activity from unauthorized or malicious behavior.

The purpose of this investigation was to analyze the Task #1003 packet capture (PCAP) using Wireshark to determine whether indicators of compromise (IoCs) or evidence of malicious activity were present within the captured network traffic.

2. Investigation Objectives

The objectives of this investigation were to:

- Analyze the Task #1003 packet capture using Wireshark.
- Establish a baseline of network traffic and protocol distribution.
- Examine DNS, TLS, QUIC, SMB, and HTTP communications.
- Identify potential indicators of compromise (IoCs) and suspicious network behavior.
- Perform packet-level inspection to evaluate potentially abnormal traffic.
- Validate suspicious domains and network indicators using external threat intelligence resources, including VirusTotal.
- Document observations, findings, and recommendations based on the completed investigation.

3. Investigation Methodology

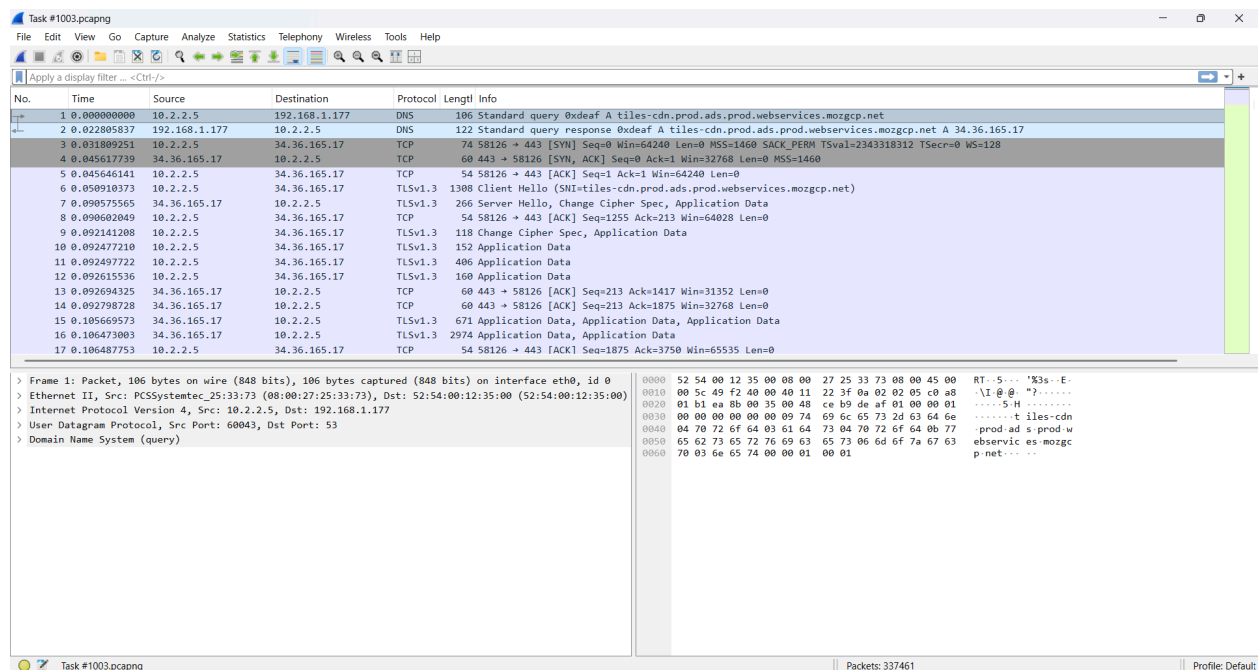
A structured methodology was followed throughout the investigation to ensure the packet capture was analyzed systematically and that conclusions were supported by evidence. The investigation consisted of four phases: preparing the packet capture for analysis, establishing a baseline of network activity, performing protocol analysis and packet inspection, and validating findings using external threat intelligence resources.

3.1 PCAP Setup

The investigation began by loading the Task #1003 packet capture (PCAP) into Wireshark. Importing the capture provided visibility into all recorded network communications and established the foundation for the investigation. The packet list was reviewed to verify that the capture loaded successfully and to identify the types of traffic contained within the dataset.

Display filters were applied throughout the investigation to isolate specific protocols and reduce unrelated traffic. Filtering allowed each protocol to be examined individually while maintaining awareness of how the communications related to the overall network activity.

Figure 1. Task #1003 packet capture loaded into Wireshark.



Observation

The packet capture loaded successfully and contained a representative sample of network communications suitable for further investigation.

Recommendation

Always verify that a packet capture has loaded correctly before beginning analysis. Establishing an organized workflow and using protocol-specific display filters improves investigative efficiency and helps ensure important traffic is not overlooked.

3.2 Baseline Traffic Analysis

Before examining individual protocols, an initial baseline of network activity was established using Wireshark's Protocol Hierarchy Statistics. Creating a baseline provides context for the investigation by identifying which protocols are present, their relative frequency, and the overall composition of network traffic.

Establishing this baseline helps analysts recognize expected communication patterns before focusing on specific protocols that may require additional investigation. This approach reduces the likelihood of misinterpreting legitimate network activity as malicious and provides direction for subsequent packet analysis.

The baseline established during this investigation guided the deeper examination of DNS, TLS, QUIC, SMB, and HTTP communications discussed in later sections of this report.

3.3 Deep Packet Inspection

Following the establishment of a baseline network profile, deeper packet inspection was performed to analyze specific protocols and communication patterns identified during the initial review. Deep packet inspection allows security analysts to examine individual packets, protocol behaviors, and communication exchanges in greater detail to identify potential indicators of compromise (IoCs) or abnormal activity.

Wireshark display filters were used to isolate specific protocols of interest, including Domain Name System (DNS), Transport Layer Security (TLS), Quick UDP Internet Connections (QUIC), Server Message Block (SMB), and Hypertext Transfer Protocol (HTTP). Each protocol was reviewed based on its role within network communication and its potential relevance during security investigations.

The analysis focused on identifying behaviors commonly associated with security threats, including suspicious domain resolution, unauthorized communication attempts, unusual encrypted traffic patterns, abnormal file-sharing activity, and unexpected data transfers. However, findings were evaluated using an evidence-based approach, recognizing that unusual traffic does not automatically indicate malicious activity.

During this phase, potentially suspicious observations were documented and investigated further through packet review and external validation methods. This approach helped differentiate legitimate network behavior from activity that could represent a security concern.

Observation

Deep packet inspection provided additional visibility into individual network communications and allowed potentially abnormal activity to be evaluated in context rather than based solely on appearance.

Recommendation

Security analysts should combine packet-level analysis with contextual investigation and threat intelligence validation before classifying network activity as malicious. This reduces false positives and supports accurate incident response decisions.

3.4 Verification Process

After identifying traffic requiring additional review, verification procedures were performed to determine whether observed activity represented legitimate network behavior or potential malicious activity.

Threat intelligence resources were used to validate indicators identified during packet analysis, including domain names and associated network communications. The domain **tiles-cdn.prod.ads.prod.webservices.mozgcp.net** was reviewed using VirusTotal and reputation-based analysis tools to determine whether the communication was associated with known malicious infrastructure.

The validation process indicated that the domain was associated with legitimate services rather than known malicious activity. This step was critical because security investigations require analysts to avoid making conclusions based solely on initial observations. A domain, IP address, or communication pattern may appear suspicious while still being part of normal business operations or trusted third-party services.

By combining Wireshark analysis with external verification resources, the investigation was able to determine whether suspicious-looking activity represented an actual security concern.

Observation

The verification process demonstrated that the identified network activity did not contain confirmed indicators of compromise. The reviewed domain was determined to be legitimate.

Recommendation

Analysts should always validate suspicious findings using reliable threat intelligence sources before escalating activity as a security incident. Proper validation helps prevent unnecessary investigations and ensures security resources are focused on genuine threats.

4. Network Traffic Analysis

Following the initial investigation methodology, detailed network traffic analysis was conducted to evaluate protocol activity, communication patterns, and potential indicators of compromise (IoCs) within the Task #1003 packet capture.

The analysis focused on establishing a baseline of normal network behavior, identifying traffic requiring additional review, and determining whether observed activity represented a potential security concern.

Wireshark was utilized throughout the investigation to review protocol statistics, apply display filters, inspect packet-level details, and analyze communication patterns between hosts.

The following sections document the analysis performed on the primary protocols and traffic patterns observed during the investigation.

4.1 Protocol Hierarchy Analysis

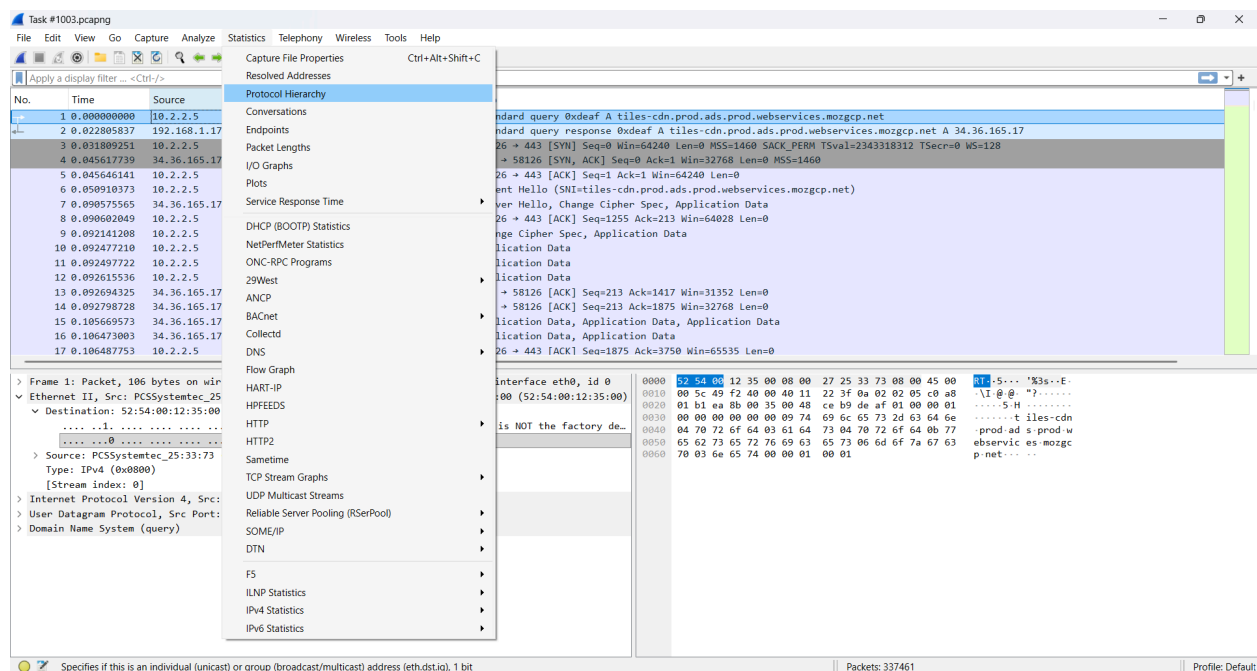
The first stage of detailed traffic analysis involved reviewing Wireshark's Protocol Hierarchy statistics to establish an overview of the protocols present within the packet capture and their distribution across network communications.

Reviewing protocol distribution is an important initial step during network investigations because it provides visibility into the communication environment before analyzing individual packets.

Understanding the protocols present and their frequency allows analysts to identify areas requiring additional investigation and establish a baseline for expected network activity.

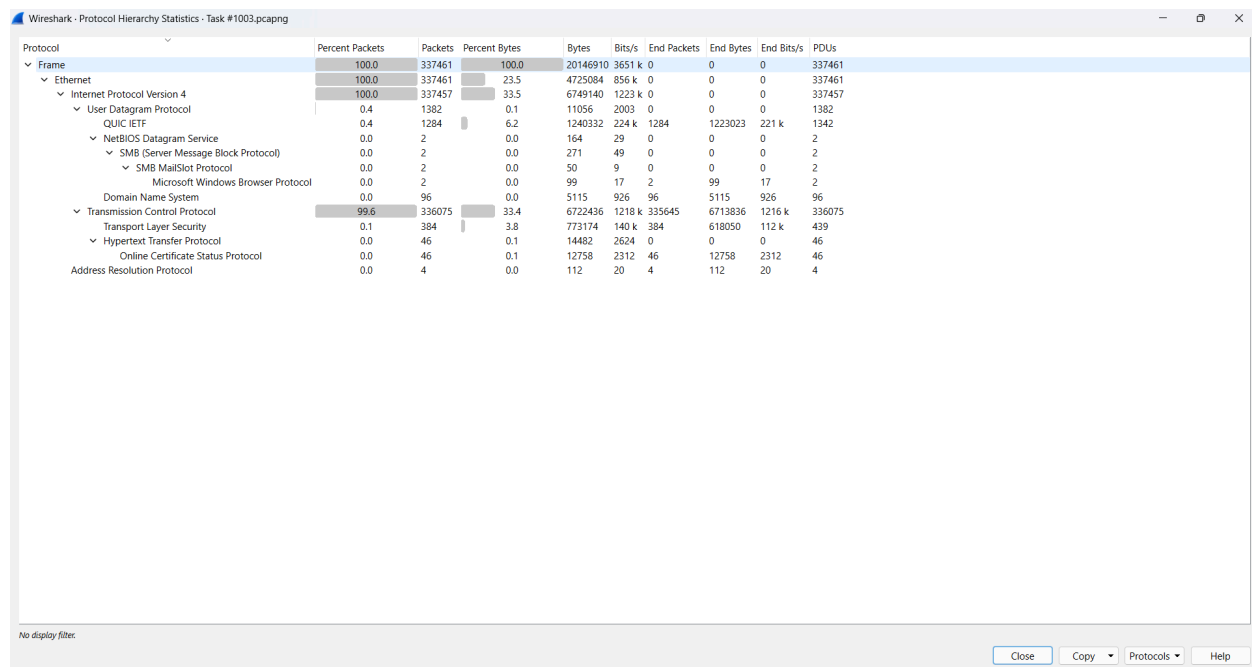
As shown in Figure 2, the Protocol Hierarchy statistics provided visibility into the types of communication contained within the Task #1003 packet capture. The captured traffic included commonly observed protocols such as DNS, TLS, QUIC, SMB, and HTTP-based communications associated with normal application and network operations.

Figure 2. Accessing Protocol Hierarchy statistics in Wireshark.



The protocol distribution was reviewed to identify unexpected, uncommon, or potentially suspicious protocols. The presence of encrypted protocols, including TLS and QUIC, was expected due to the widespread use of secure web communications and modern application services. SMB traffic was also reviewed due to its importance within enterprise environments and its potential association with unauthorized file access, credential exposure, or lateral movement during security incidents.

Figure 3. Protocol Hierarchy distribution for Task #1003 packet capture.



The Protocol Hierarchy review established an initial baseline of the captured network activity and identified areas requiring deeper analysis. Protocols were not considered suspicious solely based on their presence; instead, each protocol was evaluated based on communication patterns, host relationships, context, and supporting packet-level evidence.

Observation

The Protocol Hierarchy analysis revealed expected network protocols within the capture, including DNS, TLS, QUIC, SMB, and HTTP traffic. No unexpected or unauthorized protocols were observed during the initial review.

Recommendation

Establishing a protocol baseline should be performed at the beginning of network traffic investigations. Understanding normal communication patterns allows analysts to more accurately identify deviations that may indicate security risks or potential compromise.

4.2 DNS Analysis

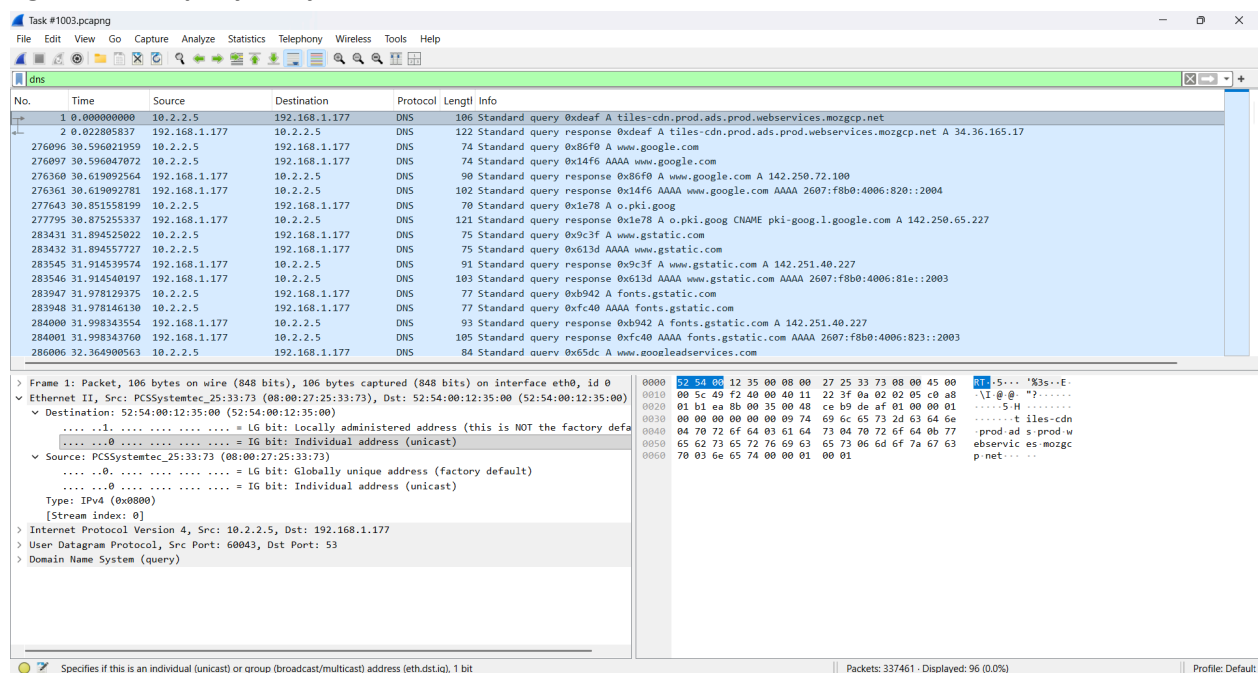
Domain Name System (DNS) analysis was conducted to evaluate domain resolution activity within the Task #1003 packet capture. DNS is a foundational network service responsible for translating human-readable domain names into corresponding IP addresses.

Because threat actors may abuse DNS infrastructure for command-and-control (C2) communication, malware callbacks, phishing operations, or DNS tunneling, DNS traffic is an important area of review during security investigations.

Wireshark display filters were applied to isolate DNS queries and responses within the packet capture. The analysis focused on identifying unusual domain requests, unexpected query patterns, failed resolution attempts, and communication with potentially suspicious infrastructure.

As shown in Figure 4, DNS traffic was reviewed to examine requested domains and the corresponding responses returned by DNS servers.

Figure 4. DNS query analysis within Wireshark.



The observed DNS requests were evaluated for characteristics commonly associated with malicious activity, including:

- Requests to suspicious or previously unknown domains.
- Excessive DNS queries that may indicate automated malware communication.
- Randomized domain structures associated with domain generation algorithms (DGAs).
- Repeated failed DNS resolution attempts.
- Communication patterns consistent with DNS tunneling.

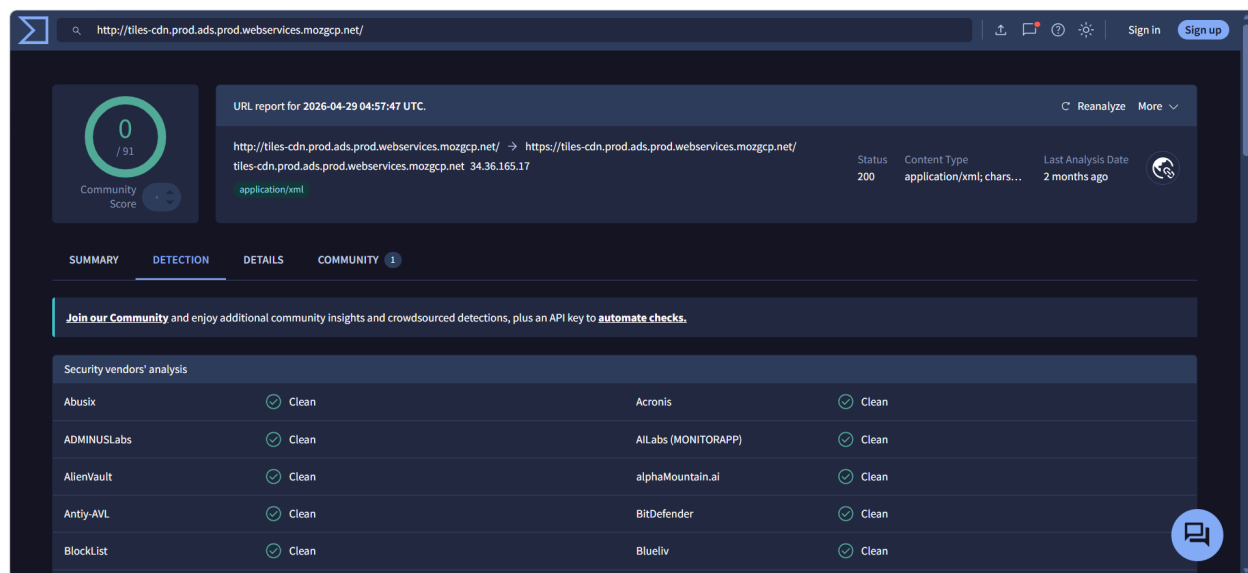
The DNS activity observed during this investigation was consistent with expected application and web-service communication patterns. The identified queries demonstrated normal resolution behavior, and no evidence of suspicious domain activity or known malicious DNS infrastructure was observed.

One domain required additional review:

tiles-cdn.prod.ads.prod.webservices.mozgcp.net

Because unfamiliar domains may appear during packet analysis, additional validation was performed using external reputation resources. The domain was reviewed through VirusTotal and reputation-based analysis tools to determine whether it demonstrated characteristics associated with malicious infrastructure.

Figure 5: VirusTotal domain reputation analysis for suspicious DNS indicator



External reputation validation indicated that the domain was associated with legitimate service activity and that no known malicious associations were identified.

Observation

DNS analysis identified normal domain resolution activity within the packet capture. Although one domain required additional review due to its unfamiliar naming structure, external validation indicated that the communication was consistent with legitimate service activity. No indicators of compromise were identified through DNS analysis.

Recommendation

Security analysts should investigate unfamiliar DNS requests rather than immediately classify them as malicious. Combining packet analysis with threat intelligence validation helps reduce false positives while ensuring potentially harmful activity is properly evaluated.

4.3 TLS Analysis

Transport Layer Security (TLS) traffic was analyzed to evaluate encrypted communications identified within the Task #1003 packet capture. TLS is a cryptographic protocol designed to secure communication between clients and servers by providing encryption, authentication, and data integrity.

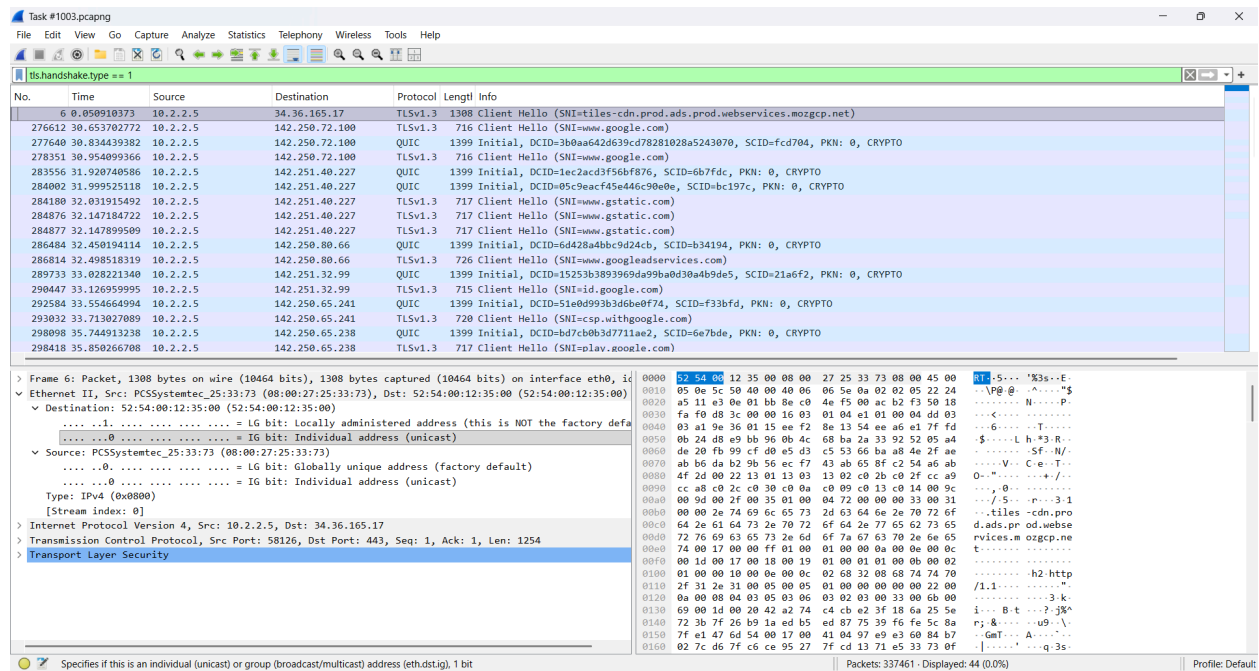
Because encryption limits visibility into the contents of network communications, TLS analysis is an important component of network traffic investigations. Although encrypted traffic is a standard security practice and does not inherently indicate malicious activity, threat actors may attempt to leverage encrypted channels to conceal command-and-control (C2) communication, data exfiltration, or unauthorized activity.

For this reason, TLS traffic must be evaluated using available metadata, including communication endpoints, certificate details, connection behavior, destination reputation, and supporting packet-level evidence.

Wireshark was used to examine TLS-related activity within the packet capture, including TLS handshake information, communicating hosts, and associated network behavior. The analysis focused on identifying potential indicators of suspicious activity, including invalid or unusual certificates, unexpected destinations, abnormal connection patterns, or encrypted communications inconsistent with expected network operations.

The TLS handshake was reviewed because it provides valuable information even when application data remains encrypted. During the handshake process, systems exchange information required to establish secure communication, including supported TLS versions, cryptographic parameters, and certificate details.

Figure 6. TLS traffic analysis within Wireshark.



The TLS communications observed within the Task #1003 packet capture demonstrated characteristics consistent with expected encrypted network activity. The reviewed connections aligned with normal application and web-service communication patterns and did not display indicators commonly associated with malicious activity conducted over encrypted channels.

The analysis did not identify suspicious certificate information, unauthorized encrypted connections, or abnormal TLS communication patterns. The observed TLS activity was consistent with modern network environments where applications and web services routinely use HTTPS and other encrypted protocols to protect data transmission.

Observation

TLS analysis identified encrypted communications consistent with expected secure network activity. No suspicious certificates, abnormal connection behavior, or indicators of compromise were identified during the review.

Recommendation

Security teams should monitor encrypted traffic while recognizing that encryption alone does not indicate a compromise. Analysts should evaluate TLS communications using available metadata, certificate details, destination reputation, and behavioral analysis to identify potentially malicious activity while minimizing false positives.

4.4 QUIC Analysis

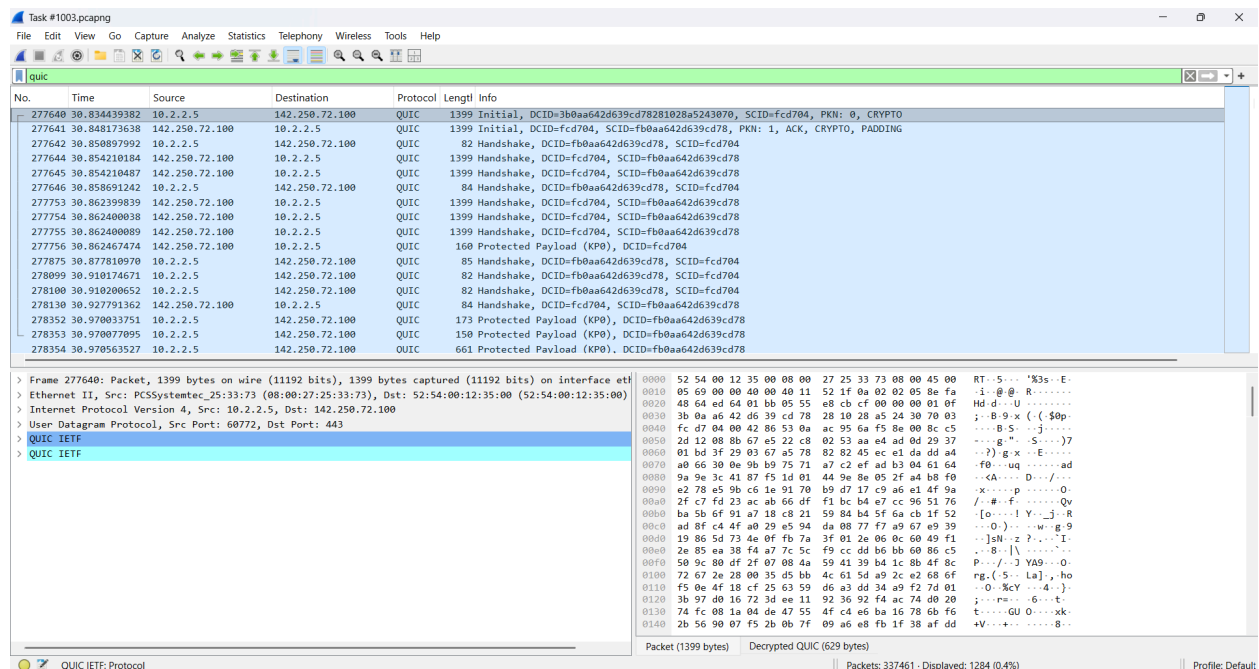
QUIC (Quick UDP Internet Connections) traffic was analyzed to evaluate encrypted UDP-based communications identified within the Task #1003 packet capture. QUIC is a modern transport protocol designed to improve the performance, reliability, and security of internet communications by combining features traditionally associated with TCP and TLS.

Unlike traditional web traffic that commonly relies on TCP-based connections, QUIC operates over the User Datagram Protocol (UDP). It is widely adopted by modern applications and web services because it supports faster connection establishment, improved performance during network changes, and integrated encryption through TLS 1.3.

From a security analysis perspective, QUIC traffic can present visibility challenges because encryption limits access to communication payloads. Additionally, because QUIC operates over UDP, analysts must evaluate traffic context, communication patterns, destination information, and available metadata to distinguish legitimate application activity from potentially suspicious behavior.

Wireshark was used to examine QUIC-related activity within the packet capture, including communication endpoints, packet behavior, and associated network patterns. The analysis focused on identifying potential indicators of suspicious activity, including unexpected destinations, abnormal connection behavior, unusual traffic patterns, or activity inconsistent with expected application behavior.

Figure 7. QUIC traffic analysis within Wireshark.



The QUIC communications observed within the Task #1003 packet capture demonstrated characteristics consistent with expected encrypted application traffic. The identified connections aligned with normal

web-service and application activity and did not demonstrate indicators commonly associated with malicious activity conducted over QUIC.

The analysis did not identify suspicious QUIC destinations, abnormal connection behavior, or unusual traffic patterns. The presence of encrypted UDP-based traffic was consistent with modern network operations, where applications increasingly utilize QUIC to improve performance and provide secure communications.

Observation

QUIC analysis identified encrypted UDP-based communications consistent with expected modern application activity. No suspicious destinations, abnormal connection behavior, or indicators of compromise were identified during the review.

Recommendation

Security teams should include QUIC traffic within routine network monitoring and investigations while recognizing that encrypted protocols are not inherently malicious. Analysts should evaluate QUIC communications using available metadata, destination reputation, traffic patterns, and behavioral indicators to identify potential threats while minimizing false positives.

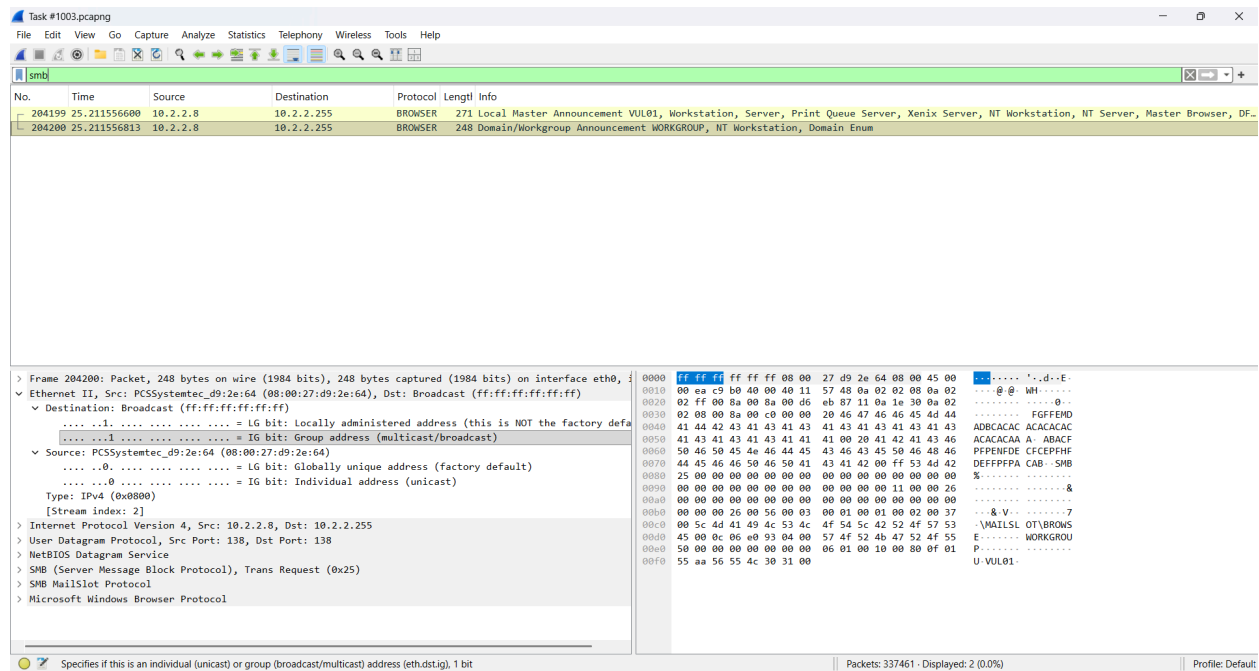
4.5 SMB Analysis

Server Message Block (SMB) traffic was analyzed to evaluate internal network communications involving file sharing, resource access, and potential indicators of unauthorized activity within the Task #1003 packet capture.

SMB is a network protocol commonly used within enterprise environments to provide access to shared files, printers, and other network resources. Because SMB supports many legitimate administrative and business functions, the presence of SMB traffic within a packet capture does not inherently indicate malicious activity. Instead, SMB communications must be evaluated within the context of the overall network environment, host relationships, and associated traffic behavior.

SMB requires careful analysis during security investigations because threat actors may abuse SMB services for activities such as lateral movement, unauthorized file access, credential theft, and malware propagation. Reviewing SMB activity allows security analysts to determine whether network resource access appears consistent with expected operations or contains characteristics associated with compromise.

Wireshark display filters were applied to isolate SMB-related traffic within the packet capture. The analysis focused on SMB endpoints, request and response activity, file-sharing behavior, and potential indicators of suspicious activity, including unauthorized access attempts, unusual resource requests, unexpected host-to-host communication, or abnormal file transfer behavior.

Figure 8. SMB traffic analysis within Wireshark.

The SMB traffic observed within the Task #1003 packet capture was reviewed in relation to surrounding network activity. The identified communications demonstrated characteristics consistent with expected internal resource access and normal file-sharing operations. No evidence of unauthorized access attempts, suspicious file transfers, or abnormal SMB behavior was identified.

The investigation also evaluated whether SMB traffic contained indicators commonly associated with lateral movement, including unexpected host communication, unusual access requests, or unauthorized resource activity. The reviewed SMB communications did not demonstrate characteristics associated with malicious SMB usage or attacker movement within the environment.

Observation

SMB analysis identified internal file-sharing activity consistent with expected network communication patterns. No evidence of unauthorized access attempts, suspicious file transfers, or lateral movement activity was observed within the packet capture.

Recommendation

Organizations should continuously monitor SMB traffic as part of routine security operations because abnormal SMB behavior may indicate unauthorized access or attacker movement within a network. Security teams should maintain secure SMB configurations, restrict unnecessary access, disable outdated SMB versions, and investigate unexpected communication patterns between hosts.

4.6 HTTP Analysis

Hypertext Transfer Protocol (HTTP) traffic was analyzed to evaluate web-based communications identified within the Task #1003 packet capture. HTTP is an application-layer protocol commonly used for communication between web clients and servers, enabling the transfer of web pages, application data, and other resources.

Although HTTP is widely used for legitimate web communication, it requires careful analysis during security investigations because threat actors may abuse web-based channels for activities such as command-and-control (C2) communication, malicious file delivery, data exfiltration, phishing operations, or distribution of unauthorized content.

Unlike HTTPS traffic, HTTP communications are transmitted without encryption, allowing analysts to directly inspect request and response information. This visibility provides valuable investigative data, including requested resources, user-agent information, server responses, status codes, and potential indicators of suspicious activity.

Wireshark display filters were applied to isolate HTTP-related communications within the packet capture. The analysis focused on HTTP requests and responses, communicating endpoints, requested resources, response codes, and potential indicators of malicious behavior, including unusual URLs, suspicious file transfers, unexpected destinations, or abnormal communication patterns.

Figure 9. HTTP traffic analysis within Wireshark.

The screenshot displays the Wireshark network protocol analyzer interface. The top menu bar includes File, Edit, View, Go, Capture, Analyze, Statistics, Telephony, Wireless, Tools, and Help. The toolbar contains icons for various functions like opening files, saving, and filtering. The main window is divided into three panes:

- Packet List:** Shows a list of captured packets. The selected packet is 24, which is an HTTP GET request from 10.2.2.5 to 34.36.165.17. The list includes columns for No., Time, Source, Destination, Protocol, Length, and Info.
- Packet Details:** Shows the hierarchical structure of the selected packet. It includes Ethernet II (Type: IPv4), Internet Protocol Version 4 (Src: 10.2.2.5, Dst: 34.36.165.17), and Transmission Control Protocol (Src Port: 443, Dst Port: 58126, Seq: 14299, Ack: 1875, Len: 1429). It also shows the reassembled TCP segments and the application data.
- Packet Bytes:** Shows the raw data of the selected packet in hexadecimal and ASCII. The data starts with the Ethernet II header and the IP header.

The status bar at the bottom indicates that the display filter is 'frame.len > 1000', showing 337461 packets, with 1075 packets displayed (0.33%).

The HTTP traffic observed within the Task #1003 packet capture was reviewed in relation to surrounding network activity. The identified communications demonstrated characteristics consistent with normal

web-based application usage and expected network operations. No evidence of suspicious HTTP requests, malicious file downloads, unauthorized data transfers, or abnormal web communication patterns was identified.

The investigation also evaluated HTTP traffic for indicators commonly associated with web-based attacks, including unexpected outbound connections, suspicious resource requests, and communication with potentially malicious infrastructure. The reviewed HTTP communications did not demonstrate characteristics associated with malicious activity or compromise.

Observation

HTTP analysis identified web-based communications consistent with expected application and network activity. No suspicious requests, malicious downloads, unauthorized data transfers, or indicators of compromise were identified within the packet capture.

Recommendation

Organizations should monitor HTTP traffic as part of routine security operations because unsecured web communications may expose sensitive information or create opportunities for malicious activity. Security teams should promote the use of HTTPS, monitor unusual web requests, analyze suspicious destinations, and investigate abnormal HTTP communication patterns.

4.7 Network Traffic Analysis Summary

Following the review of individual protocols within the Task #1003 packet capture, an overall assessment of network activity was conducted to determine whether any indicators of compromise (IoCs), suspicious communication patterns, or abnormal behaviors were present.

The analysis included reviewing protocol distribution, DNS activity, encrypted communications, internal network traffic, and web-based communications using Wireshark. Each protocol was evaluated based on communication behavior, host relationships, destination information, and supporting packet-level evidence rather than protocol presence alone.

The Protocol Hierarchy analysis established a baseline of expected network activity and identified commonly observed protocols, including DNS, TLS, QUIC, SMB, and HTTP. Further analysis of these protocols did not identify unusual services, unauthorized communications, or unexpected protocol usage.

DNS analysis identified normal domain resolution activity associated with expected application and web-service communications. Although one domain required additional validation due to its unfamiliar naming structure, external reputation analysis confirmed that the communication was associated with legitimate service activity. No evidence of malicious DNS behavior, DNS tunneling, or suspicious domain resolution patterns was identified.

Encrypted communication analysis involving TLS and QUIC identified expected secure application traffic. The reviewed connections did not demonstrate suspicious certificate activity, abnormal communication patterns, or unexpected destinations. The presence of encrypted protocols was consistent with modern network environments where applications rely on secure communication methods to protect data transmission.

SMB analysis identified internal file-sharing activity consistent with expected network operations. No evidence of unauthorized access attempts, suspicious file transfers, credential-related abuse, or lateral movement activity was observed within the packet capture.

HTTP analysis identified normal web-based communications and application activity. The review did not identify suspicious requests, malicious file downloads, unauthorized data transfers, or communication with known malicious infrastructure.

Based on the combined protocol analysis, the Task #1003 packet capture did not reveal evidence of active compromise, malicious network activity, or unauthorized communication. The observed traffic patterns were consistent with normal application behavior and expected network operations.

Observation

The comprehensive network traffic analysis identified expected communication patterns across the reviewed protocols. No indicators of compromise, suspicious network behavior, or unauthorized activity were identified within the Task #1003 packet capture.

Recommendation

Organizations should continue performing network traffic analysis as part of routine security monitoring and incident response processes. Establishing communication baselines, reviewing protocol behavior, validating suspicious findings through threat intelligence resources, and analyzing traffic patterns over time can improve the detection of abnormal activity and potential security threats.

5. Findings and Analysis

Following the completion of network traffic analysis, the findings from the Task #1003 packet capture were reviewed to determine whether any security concerns, indicators of compromise (IoCs), or abnormal network behaviors were present.

The investigation evaluated multiple areas of network activity, including protocol distribution, DNS resolution behavior, encrypted communications, internal network traffic, and web-based communications. Each finding was assessed based on observed communication patterns, host relationships, destination information, and supporting packet-level evidence.

The analysis identified network activity consistent with expected application and system operations. Protocols observed within the packet capture, including DNS, TLS, QUIC, SMB, and HTTP, demonstrated communication patterns consistent with legitimate network activity.

During the investigation, one DNS entry required additional review due to an unfamiliar domain structure:

tiles-cdn.prod.ads.prod.webservices.mozgcp.net

Additional validation was performed using reputation-based analysis tools to determine whether the domain demonstrated characteristics associated with malicious infrastructure. The review confirmed that the domain was associated with legitimate service activity and did not demonstrate indicators commonly associated with malicious behavior.

Encrypted communications identified through TLS and QUIC analysis were reviewed due to the reduced visibility associated with encrypted traffic. The investigation did not identify suspicious certificates, abnormal connection patterns, unauthorized encrypted communications, or communication with destinations associated with known malicious activity.

SMB traffic was evaluated due to its importance within enterprise environments and its potential association with lateral movement, unauthorized access, credential exposure, and malware propagation. The observed SMB communications were consistent with expected internal resource access, and no evidence of malicious SMB activity was identified.

HTTP communications were reviewed for indicators commonly associated with web-based threats, including suspicious requests, malicious downloads, unauthorized data transfers, or communication with potentially harmful infrastructure. The analysis did not identify abnormal HTTP activity or evidence of web-based compromise.

Based on the collected evidence, the Task #1003 packet capture did not demonstrate signs of compromise, unauthorized access, or malicious network activity. The observed traffic patterns aligned with expected application behavior and normal network operations.

Finding Summary

Finding Area	Result
Protocol Analysis	Expected protocols identified; no suspicious or unauthorized protocols observed
DNS Activity	Legitimate domain resolution activity identified; no malicious DNS behavior detected
TLS Communications	Expected encrypted traffic observed; no certificate issues or connection anomalies identified
QUIC Traffic	Normal encrypted UDP communication observed; no suspicious behavior identified
SMB Activity	Expected internal file-sharing activity observed; no indicators of lateral movement identified
HTTP Traffic	Normal web communication observed; no malicious requests, downloads, or transfers identified

Overall Assessment

Based on the evidence collected during the investigation, the Task #1003 packet capture did not contain indicators of compromise (IoCs), malicious network activity, or unauthorized communication. The observed network traffic was consistent with expected application behavior and normal network operations. Accordingly, the packet capture was assessed as containing no evidence of an active security incident.

Recommendation

Organizations should continue monitoring network traffic through established security monitoring and incident response processes. Security teams should maintain documented network communication baselines, regularly review anomalous traffic patterns, validate suspicious findings using trusted threat intelligence resources, and investigate deviations that may indicate potential security incidents.

6. Security Recommendations

Although the Task #1003 packet capture analysis did not identify confirmed malicious activity, the investigation reinforced several security practices that organizations should maintain to support effective threat detection, monitoring, and incident response.

The following recommendations are based on observations made during the network traffic analysis process.

6.1 Maintain Continuous Network Monitoring

Organizations should implement continuous network monitoring to identify changes in normal traffic patterns and detect potential security concerns as they occur.

Establishing a baseline of expected network behavior allows security teams to identify anomalies, including unusual communication patterns, unexpected protocol usage, unauthorized connections, or changes in normal host behavior.

Security monitoring solutions, including Security Information and Event Management (SIEM) platforms and network monitoring tools, should be used to collect, analyze, and correlate network activity across the environment.

Recommendation

Maintain ongoing visibility into network traffic and establish documented baselines for normal communication patterns. Consistent monitoring improves an organization's ability to detect, investigate, and respond to potential threats before they develop into larger security incidents.

6.2 Validate Suspicious Indicators Before Escalation

The investigation demonstrated the importance of validating potentially suspicious activity before determining whether it represents a security threat.

The domain **tiles-cdn.prod.ads.prod.webservices.mozgcp.net** initially required additional review due to its unfamiliar naming structure. However, external threat intelligence validation confirmed that the domain was associated with legitimate service activity.

Security analysts should avoid relying solely on initial observations when evaluating potential threats. Proper validation helps reduce false positives, prevents unnecessary disruption of legitimate services, and ensures investigative resources are focused on credible security risks.

Recommendation

Use multiple sources of evidence, including packet analysis, threat intelligence platforms, domain reputation services, and organizational context before escalating suspected indicators.

6.3 Monitor Encrypted Network Traffic

Modern applications rely heavily on encrypted communication protocols such as TLS and QUIC. While encryption protects confidentiality and data integrity, it can reduce visibility into network activity.

Security teams should maintain appropriate monitoring capabilities for encrypted communications by analyzing available metadata, including connection behavior, certificate information, destinations, traffic patterns, and communication frequency.

Recommendation

Implement monitoring strategies that evaluate encrypted traffic without treating encryption itself as suspicious. Analysts should focus on behavioral indicators and communication context rather than protocol usage alone.

6.4 Maintain Secure SMB Configurations

SMB remains an important protocol within enterprise environments but can introduce security risks when improperly configured. Attackers may target SMB services for unauthorized access, credential theft, lateral movement, and malware propagation.

Organizations should ensure SMB services are securely configured by limiting unnecessary access, applying security updates, disabling outdated SMB versions, and monitoring unusual communication patterns.

Recommendation

Regularly review SMB configurations and restrict access according to business requirements. Continuous monitoring of SMB activity can help identify unauthorized access attempts and potential adversary movement within the network.

6.5 Perform Regular Network Traffic Analysis

Periodic packet analysis provides valuable insight into network behavior and helps security teams identify potential risks that may not be detected through automated monitoring solutions alone.

Manual analysis using tools such as Wireshark remains an important skill for cybersecurity professionals because it provides detailed visibility into network communications and supports incident investigation activities.

Recommendation

Include network traffic analysis as part of routine security operations, incident response exercises, and troubleshooting procedures. Combining automated monitoring with analyst-driven investigation improves overall security visibility and threat detection capabilities.

Summary of Recommendations

The Task #1003 investigation did not identify an active security threat; however, maintaining strong security practices remains essential for reducing risk and improving detection capabilities.

Organizations should continue monitoring network activity, validating suspicious indicators, securing internal protocols, and maintaining visibility into communications across their environments. A combination of automated security monitoring, threat intelligence validation, and analyst-driven investigation provides a stronger foundation for identifying and responding to potential security incidents.

7. Conclusion

The Task #1003 Wireshark-based packet analysis investigation was conducted to evaluate captured network traffic and determine whether indicators of compromise (IoCs), suspicious communication patterns, or evidence of malicious activity were present. A structured investigation methodology was followed, beginning with establishing a baseline of expected network activity and progressing through protocol analysis, packet inspection, communication analysis, and threat intelligence validation.

The investigation examined multiple areas of network communication, including protocol distribution, DNS queries, TLS-encrypted traffic, QUIC communications, SMB activity, HTTP communications, packet behavior, and domain reputation. Wireshark provided detailed visibility into the captured traffic, allowing network communications and behavioral patterns to be evaluated within their operational context.

During the investigation, certain activity required additional review due to characteristics that could potentially appear suspicious. However, through packet-level analysis and external validation using threat intelligence resources, the reviewed activity was determined to be legitimate. The domain **tiles-cdn.prod.ads.prod.webservices.mozgcp.net**, which required additional verification, was confirmed to be associated with legitimate service activity rather than malicious infrastructure.

The investigation did not identify confirmed indicators of compromise, unauthorized access attempts, suspicious data transfers, abnormal network behavior, or communication with known malicious infrastructure. The observed traffic patterns were consistent with expected application activity and normal network operations.

A key takeaway from this investigation is the importance of evidence-based decision-making during security assessments. Cybersecurity analysts must avoid drawing conclusions based solely on individual indicators, such as unfamiliar domains, encrypted traffic, or unexpected protocol usage. Instead, findings should be evaluated using multiple sources of evidence, including packet analysis, behavioral analysis, environmental context, and threat intelligence validation.

This investigation demonstrated the importance of a structured network analysis process in distinguishing legitimate activity from potential security threats. By applying Wireshark analysis techniques and following a methodical investigative approach, the captured traffic was accurately assessed and determined not to represent a security incident.

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